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Nutrition Chatbot Using DeepSeek-R1 Local LLM: A Case Study on Healthy Eating Habits

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ABSTRACT This report presents the development and testing of a chatbot specialized in nutrition and healthy eating habits. The system uses DeepSeek-R1 7B, a local open-source language model running through Ollama, combined with a structured knowledge base containing nutritional information for 28+ foods, meal recommendations, and health tips. The frontend was developed using Streamlit, providing a chat-style interface. The chatbot was evaluated using 10 validation questions covering daily servings, vitamin C sources, water intake, protein requirements, cholesterol, dietary comparisons, thyroid conditions, weight loss, carbohydrates, and digestion. Results indicate that the chatbot answers all questions correctly, with response times ranging from 30 seconds for simple questions to 120 seconds for complex questions. Identified limitations include high latency due to local hardware constraints, memory consumption (8GB RAM), session degradation requiring page refresh after 3-4 questions, and language mixing issues in early tests. Proposed improvements include streaming mode, automatic session refresh, conversational memory, and GPU deployment.

INDEX TERMS Chatbot, nutrition, DeepSeek, large language models, prompt engineering, Streamlit, healthy eating, Ollama.

I. INTRODUCTION

NUTRITION plays a fundamental role in human health and well-being. However, many people lack access to reliable, personalized nutritional information. Chatbots powered by large language models (LLMs) offer a promising solution to provide accessible, interactive nutritional guidance.

The transformer architecture introduced by Vaswani et al. [1] revolutionized natural language processing and enabled the development of large language models. Subsequent work by Brown et al. [2] and Radford et al. [3] demonstrated the capabilities of LLMs in understanding and generating human-like text.

This report presents the development of a chatbot specialized in nutrition and healthy eating habits. The system was built using DeepSeek-R1 7B [4], a local open-source LLM running through Ollama [5], combined with a structured knowledge base containing nutritional information. The frontend was developed using Streamlit [6], a Python framework for rapid web application development.

The main contributions of this work are:

1) Development of a functional nutrition chatbot using a local open-source LLM (DeepSeek-R1 7B)
2) Integration of structured nutritional knowledge with prompt engineering techniques
3) Web-based user interface using Streamlit with support for Spanish and English questions
4) Evaluation using 10 validation questions covering key nutrition topics
5) Identification of limitations and proposal of improvements for future versions

II. RELATED WORK

A. LARGE LANGUAGE MODELS IN HEALTHCARE

Recent systematic reviews by Thirunavukarasu et al. [7] and Chung et al. [8] have examined the applications of LLMs in medical settings. These studies demonstrate that language models can provide useful information in medical domains, although challenges remain in terms of accuracy and reliability.

B. LOCAL LLMS FOR PRIVACY-PRESERVING APPLICATIONS

Wang et al. [9] highlighted the importance of local LLMs for privacy-preserving healthcare applications. One important advantage of using local LLMs such as DeepSeek and Llama is that they can run on local hardware without sending sensitive data to external servers. This is particularly relevant in healthcare applications where patient privacy is critical.

C. CHATBOTS FOR NUTRITION AND HEALTHY EATING

Zhang et al. [10] conducted a systematic review of AI-powered nutrition chatbots. Lee et al. [11] specifically evaluated chatbot responses for dietary advice accuracy and safety. Previous work has developed rule-based chatbots for nutrition advice, but these systems are limited by their predefined rules. More recent approaches have explored the use of LLMs to provide more flexible and natural conversational experiences.

D. CONNECTION TO OUR WORK

Based on these ideas, this project implements a nutrition chatbot using a local LLM (DeepSeek-R1 7B) combined with a structured knowledge base. The system is designed to provide accurate nutritional information while maintaining user privacy by running entirely on local hardware. The complete implementation is available in our repository [12].

III. OBJECTIVE

The main objective of this project is to develop and evaluate a chatbot capable of answering questions related to nutrition and healthy eating habits using a local open-source LLM.

Specific objectives include:

1) Compile and model nutritional knowledge from reliable sources.
2) Implement prompt engineering techniques to guide the LLM's responses.
3) Develop a simple web interface for user interaction.
4) Evaluate the chatbot's performance using a set of 10 validation questions.
5) Identify limitations and propose improvements for future work.

IV. METHODOLOGY

A. SELECTED DOMAIN

For this project, the selected domain was **nutrition and healthy eating habits**. This domain was chosen because it is relevant to many people and contains a large amount of accessible information from reliable sources such as WHO and FAO.

B. DOMAIN DESCRIPTION

Basic human nutrition encompasses knowledge about macronutrients (carbohydrates, proteins, fats), micronutrients (vitamins and minerals), daily recommended intakes, food sources, and the relationship between diet and common health conditions. This domain has a strong evidence base from nutritional science.

C. KNOWLEDGE BASE STRUCTURE

The compiled knowledge was organized into a structured Python dictionary (`conocimiento.py`) containing:

- **Foods database**: 28+ common foods with nutritional properties (calories, proteins, carbohydrates, fats, fiber, and key benefits)
- **Meal recommendations**: Specific suggestions for breakfast, lunch, dinner, and healthy snacks
- **Nutritional comparisons**: Predefined comparisons between foods (e.g., fried chicken vs Russian salad)
- **Health tips**: General nutrition advice based on WHO guidelines
- **Condition-specific advice**: Nutritional recommendations for thyroid problems, cholesterol management, and weight loss

D. KNOWLEDGE MODELING APPROACH

The collected knowledge was modeled using the following approach:

1) Information was extracted from reliable sources (WHO, FAO, Mayo Clinic) and manually structured.
2) A Python module (`conocimiento.py`) was created containing dictionaries for foods, recommendations, comparisons, and tips.
3) Functions for searching, comparing, and retrieving relevant information were implemented.
4) The structured knowledge is injected into the LLM's prompt at runtime based on the user's question.

E. PROMPT ENGINEERING STRATEGY

The prompt engineering approach includes:

1) **Intent detection**: The system analyzes keywords related to calories, comparisons, recommendations, benefits, or general advice.
2) **Context injection**: Relevant knowledge from the structured database is dynamically inserted into the prompt.
3) **System role definition**: The LLM is instructed to act as a certified nutritionist.
4) **Response constraints**: The model is limited to brief, clear answers (4-5 lines maximum) using emojis for friendliness.
5) **Language consistency**: The system detects whether the question is in Spanish or English and responds in the same language.

F. LLM INTEGRATION

The chatbot uses **DeepSeek-R1 7B** [4], a local open-source language model running via Ollama [5]. Table 1 summarizes the technical stack used in development.

TABLE 1: Technical stack

Component	Technology
LLM	DeepSeek-R1 7B
Model runner	Ollama
Frontend	Streamlit
Language	Python 3.11
API communication	HTTP REST (port 11434)

The integration works as follows:

1) User question is received through the web interface (Streamlit)
2) The system detects intent and retrieves relevant knowledge from the structured database
3) A prompt is constructed combining system instruction, retrieved context, response rules, and the user's question
4) The prompt is sent to DeepSeek via HTTP API (`localhost:11434/api/generate`)
5) The response is returned to the user interface

G. WEB INTERFACE

The frontend was developed using **Streamlit** [6]. Figure 1 shows the web interface of the nutrition chatbot.

FIGURE 1: Web interface of the nutrition chatbot

H. VALIDATION QUESTIONS

The chatbot was evaluated using the following 10 questions, following the evaluation methodology proposed by Lee et al. [11]:

1) How many servings of fruits and vegetables are recommended daily?
2) Which foods are rich in vitamin C?
3) How much water should an average adult drink per day?
4) How much protein does a 70 kg person need per day?
5) Does egg consumption increase cholesterol?
6) Which diet is better: Mediterranean or ketogenic?
7) What to eat if I suffer from thyroid problems?
8) What foods should I never eat if I want to lose weight?
9) Are carbohydrate-based diets effective?
10) What food is ideal to have good digestion at breakfast?

V. EXPLORATORY ANALYSIS

A. DATA QUALITY ASSESSMENT

The knowledge base was analyzed to verify its completeness and correctness. The 28 foods included cover the main food groups: fruits, vegetables, proteins, grains, and dairy. Each entry includes nutritional values per 100g and a list of key benefits.

B. KNOWLEDGE DISTRIBUTION

Table 2 shows the distribution of foods by category in the knowledge base.

TABLE 2: Food distribution by category

Category	Number of foods
Fruits	6
Vegetables	5
Proteins	7
Grains & tubers	6
Dairy & alternatives	2
Nuts & seeds	2
Mixed dishes	2

VI. EXPERIMENTS

A. INPUT AND OUTPUT FEATURES

The input to the system is a natural language question from the user. The output is a text response generated by the LLM, optionally augmented with information from the structured knowledge base.

B. TEST RESULTS

The chatbot was tested with the 10 validation questions. Table 3 presents a summary of the results.

TABLE 3: Test results summary

#	Question	Resul
1	How many servings of fruits and vegetables are recommended daily?	Correc
2	Which foods are rich in vitamin C?	Correc
3	How much water should an average adult drink per day?	Correc
4	How much protein does a 70 kg person need per day?	Correc
5	Does egg consumption increase cholesterol?	Correc
6	Which diet is better: Mediterranean or ketogenic?	Correc
7	What to eat if I suffer from thyroid problems?	Correc
8	What foods should I never eat if I want to lose weight?	Correc
9	Are carbohydrate-based diets effective?	Correc
10	What food is ideal to have good digestion at breakfast?	Correc

C. PERFORMANCE ANALYSIS

During testing, the following performance characteristics were observed:

- **Response time**: The chatbot takes between 30 and 120 seconds to respond. Complex questions (diet comparisons, detailed nutritional advice) take approximately

120 seconds, while simple questions (calories, basic facts) take 30-45 seconds.

- **Response quality**: All 10 validation questions were answered correctly and completely on the first attempt.
- **Session stability**: After approximately 3 to 4 consecutive questions, the Streamlit interface requires a page refresh to continue responding properly.
- **Language handling**: Some language mixing was observed in early tests (e.g., "kaloras" instead of "calorías"). This was resolved by strengthening prompt instructions.
- **Cold start**: The first query after starting Ollama takes longer (up to 120 seconds) because the model loads into RAM.

VII. CONCLUSIONS

This project developed and evaluated a nutrition chatbot using DeepSeek-R1 7B as the underlying LLM. The system successfully answers all 10 validation questions correctly and completely on the first attempt.

Response times vary from 30 seconds for simple questions to approximately 2 minutes (120 seconds) for complex questions. A practical limitation identified was the need to refresh the Streamlit page after every 3 to 4 consecutive questions due to session state degradation.

The system demonstrates that local open-source LLMs can be effectively combined with structured knowledge injection and prompt engineering to create a functional nutrition chatbot. As noted by Wang et al. [9], local LLMs offer significant privacy advantages for healthcare applications.

VIII. LIMITATIONS AND FUTURE WORK

A. LIMITATIONS

The following limitations were identified during testing:

- **High latency**: DeepSeek-R1 7B requires 30-120 seconds per response due to local hardware constraints. This is consistent with findings reported by Thirunavukarasu et al. [7].
- **Memory consumption**: Requires approximately 8GB of free RAM.
- **Session degradation**: Page refresh needed after 3-4 consecutive questions.
- **Limited knowledge base**: Only 28 foods currently included. Zhang et al. [10] recommend expanding knowledge bases for better coverage.
- **No conversation memory**: Each question is treated independently.
- **No source attribution**: Does not provide references for information.

B. FUTURE WORK

Future improvements, informed by Lee et al. [11] recommendations, include:

- Enable streaming mode (`stream=True`) for real-time responses
- Add "Clear conversation" button to avoid manual page refresh
- Expand knowledge base to 100+ foods with complete vitamin/mineral profiles
- Implement conversational memory for follow-up questions
- Add smaller fallback model for low-resource hardware
- Deploy with GPU acceleration for sub-10 second responses, as suggested by Wang et al. [9]

C. REPOSITORY INFORMATION

The complete source code, demonstration video, and this report are available at:

https://github.com/Lin0u/Proyecto1.git

This implementation is fully documented in the GitHub repository by Lino Ramirez et al. [12].

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